

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());

    /* Create child process */
    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    /* Child process */
    if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        printf("Executing ls using exec()...\n");

        execlp("ls", "ls", "-l", NULL);

        perror("exec failed");
        return 1;
    }

    /* Parent process */
    else
    {
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        /* Wait for child */
        wait(NULL);

        printf("\nChild process completed.\n");
        printf("Parent process completed.\n");
    }

    return 0;
}
```

Output

```
[ubuntu@ubuntu:~/OSLab/Practical$ ./program4
Parent Process Started
Parent PID: 2328

Parent Process
Parent PID: 2328
Child PID: 2329

Child Process
Child PID: 2329
Parent PID: 2328
Executing ls using exec()...
total 124
-rw-rw-r-- 1 ubuntu ubuntu    0 Aug 25 11:57 2
-rwxrwxr-x 1 ubuntu ubuntu 70760 Sep 10 10:41 program2
-rw-rw-r-- 1 ubuntu ubuntu   741 Sep 10 10:41 program2.c
-rwxrwxr-x 1 ubuntu ubuntu 70784 Sep 10 10:51 program3
-rw-rw-r-- 1 ubuntu ubuntu  1077 Sep 10 10:51 program3.c
-rwxrwxr-x 1 ubuntu ubuntu 70744 Sep 10 11:05 program4
-rw-rw-r-- 1 ubuntu ubuntu   789 Sep 10 11:04 program4.c
-rwxrwxr-x 1 ubuntu ubuntu 70848 Sep 10 11:01 program5
-rw-rw-r-- 1 ubuntu ubuntu   928 Sep 10 11:01 program5.c
-rwxrwxr-x 1 ubuntu ubuntu 70896 Aug 25 11:55 program_1
-rw-rw-r-- 1 ubuntu ubuntu   908 Aug 25 11:54 program_1.c
-rwxrwxr-x 1 ubuntu ubuntu 70760 Aug  6 22:41 program_2-
-rw-rw-r-- 1 ubuntu ubuntu  1261 Aug  6 22:33 program_2.c
-rw-rw-r-- 1 ubuntu ubuntu    46 Aug  6 22:40 source

Child process completed.
Parent process completed.
ubuntu@ubuntu:~/OSLab/Practical$
```